Actuality Without Exhaustion

Opposition, Focus, and Recurrence Without Return

Latent–Actual Project

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Abstract

This paper develops a Lean-checked structural metaphysics of actuality and latency. Determinations inhabit unlabelled two-pole oppositions; presentations distinguish articulation from realization; and continuations settle a target while carrying a fresh trace of their predecessor. Under an explicit plenitude postulate, the possible is locally productive and exceeds every single selection whenever a presentation is supplied, whereas a focus defines a productive, target-determinate actual sublayer. On histories rooted in one effective seed, realized, excluded, open, and undrawn content coexist at every reachable presentation. An articulation-complete trace-encoding law marks archival precedence within later presentations as a strict past. A two-beat cadence makes each promotion followed by reversal, yet this past grows, yielding reversal – and, in the wider possible layer, restricted qualitative recurrence – without return. An absolutely null countermodel shows that the formalism does not derive actuality from presentation alone. Its achievement is therefore conditional and structural: it identifies assumptions under which determinacy, novelty, non-exhaustion, internal succession, and historical irreversibility are jointly coherent.

Keywords: actuality; latency; opposition; process metaphysics; modality; recurrence; formalization; Lean 4

Contents

1 The problem **3**

2 Formal method and interpretive discipline **3**

2.1 Why formalize a metaphysical argument? . . . 3

2.2 Three levels . . . 4

2.3 Conditionality as a result . . . 4

3 Opposition before orientation **4**

3.1 The field . . . 4

3.2 Unlabelled opposition and orientation . . . 5

4 Presentation and stratified latency **5**

4.1 Articulation, realization, and resolution . . . 5

4.2 Four status notions . . . 6

4.3 Offer and trace . . . 6

5 Continuation and the archive 7
5.1 What a continuation does . . . 7
5.2 A strict archive . . . 7
5.3 An internally marked past . . . 8
5.4 Recurrence without return . . . 9

6 Possibility, plenitude, and excess over selection 9
6.1 Generative layers . . . 9
6.2 The strength of plenitude . . . 9
6.3 Why possibility exceeds every selector . . . 10

7 Focus and determinate actuality 10
7.1 From a selector to a focus . . . 10
7.2 What determinacy amounts to . . . 11
7.3 Cadence . . . 11
7.4 The swirl . . . 11

8 Rooted history and permanent latency 12
8.1 One effective seed . . . 12
8.2 The strongest coexistence result . . . 12
8.3 The void and the limit of the anti-nullity argument . . . 13

9 The concrete Stage model 13

10 Metaphysical interpretation 14
10.1 A process ontology with local determination . . . 14
10.2 History marked within presentations . . . 14
10.3 Latency is stratified, not a second realm . . . 14
10.4 Possibility as lawful transition . . . 15
10.5 Repetition with difference . . . 15
10.6 Individuation and the seed . . . 15
10.7 Internal observation, minimally construed . . . 16

11 Objections, replies, and limits 16

12 Conclusion 18

A Argument-to-theorem map 18

B Premise ledger 20

1 The problem

Metaphysics repeatedly confronts a cluster of tensions. If actuality is determinate, how can it remain open to more than it presently is? If change preserves enough structure to count as continuation, how can it be genuinely novel? If a quality returns, must the world itself return? And if several continuations are possible, what distinguishes the one that is actual without reducing possibility to a disguised list of actual facts?

The distinction between potentiality and actuality is ancient, but the present account is not an exegesis of Aristotle's *Metaphysics* [1] or of any later system. It isolates a narrower structural question:

> Can one give a consistent, explicit account in which actuality is locally determinate and generative, latency remains permanent, qualitative reversal coexists with historical advance, and the possible exceeds every single actual orientation?

The answer formalized here is conditional but positive. The conditions are:

(i) a family of binary, unlabelled oppositions;

(ii) presentations that distinguish articulation from realization;

(iii) continuations that settle a target, preserve old articulation, and carry a fresh trace;

(iv) an optional plenitude principle for lawful continuations;

(v) a total focus that selects the target of actual transitions; and

(vi) one realized seed at the root of the history under consideration.

From these premises the formal development proves local productivity, target determinacy, permanent latent excess, a strict archive order, finite no-return, cadence parity, and the swirl: every promotion is followed by realization of the promoted determination's contrary while the archive moves on.

This result has a deliberately modest modal force. It is not a proof that the world has the proposed structure. It is not a derivation of being from non-being. It is a representation theorem: if the primitives and transition laws have the stated form, then a distinctive package of metaphysical consequences follows. The Lean formalization makes the conditional character useful rather than evasive, because it records exactly which conclusions use which premises and supplies countermodels where stronger conclusions fail.

2 Formal method and interpretive discipline

2.1 Why formalize a metaphysical argument?

Formalization cannot decide whether a metaphysical primitive is adequate to experience or nature. It can, however, perform three kinds of conceptual work. First, it prevents equivocation between nearby notions. Here *realized*, *articulated*, *open*, and *possible* have different types and proof conditions. Second, it exposes hidden premises. No-return, for example, is not obtained from bare opposition; it depends on fresh traces, their carriage, and monotone articulation. Third, it separates implication from interpretation. Lean verifies the implication. The metaphysical reading remains an argument whose scope can be assessed independently.

The companion source is `LatentActual.lean`, revision 11. It imports only Lean's core `Init` module and is checked with Lean 4.24.0 [4]. The concrete model and principal regression routes contain no user-declared `axiom`, `sorry`, `admit`, or `native_decide`; executable `#print axioms` guards report only `propext` and `Quot.sound` for the checked declarations.

2.2 Three levels

The theory is easiest to understand when three levels are kept separate.

Determinations. These are poles within oppositions. Contrary determination belongs to the same opposition.

Presentations. These assign articulation and realization status to determinations and carry a seed, offer, trace, and cadence bit.

Transition layers. These relate a source presentation, successor presentation, and settled determination. The possible layer records lawful continuations; the actual layer is the part of that relation selected by a focus.

This separation blocks two tempting slides. First, the contrary operation on a determination is not propositional negation. Second, there are two technical senses near the ordinary word *actual*: $\mathsf{Real}_p(d)$ is a predicate saying that content d is realized in presentation p, while an actual transition is an edge in a focused subrelation. A realized content is not itself a transition, and an actual transition is defined through more than the realization predicate.

2.3 Conditionality as a result

The account labels its strong principles instead of hiding them. The open offer, fresh trace, and articulation-complete trace-encoding law are fields of the presentation structure. Trace carriage and cadence are fields of continuation. Plenitude is an optional system-level capability. Rooted seed effectiveness is a local historical premise. Thus the formal result does not say that opposition alone produces time, novelty, or being. It says which package of assumptions suffices, how the pieces interact, and where they do not.

3 Opposition before orientation

3.1 The field

A `Field` consists of a type O of oppositions and, for each $o \in O$, a type $P(o)$ of poles. It provides a pole exchange

$$\mathsf{flip}_o : P(o) \longrightarrow P(o)$$

with no fixed point, together with the dichotomy

$$\forall p, q \in P(o), \qquad p = q \ \vee \ p = \mathsf{flip}_o(q).$$

Equality of oppositions is decidable. A determination is a dependent pair

$$d = (\mathsf{opp}(d), \mathsf{pole}(d)), \qquad \mathsf{pole}(d) \in P(\mathsf{opp}(d)).$$

Its contrary is

$$\overline{d} = (\mathsf{opp}(d), \mathsf{flip}(\mathsf{pole}(d))).$$

The two-pole dichotomy and fixed-point freedom imply, rather than assume, that flip is involutive:

$$\overline{\overline{d}} = d, \qquad \overline{d} \neq d.$$

These are `Field.neg_involutive` and `Field.neg_free`. If two determinations share an opposition, then one is either the other or its contrary (`Field.eq_or_neg`).

Qualification. A bare `Field` does not require every $P(o)$ to be inhabited. Strictly speaking, an inhabited fibre is a free transitive two-element action, while an empty fibre satisfies the universally quantified field laws vacuously. A total selector in a full system supplies a pole at every opposition it is asked to select. Claims that all field fibres are two-pole torsors should therefore be read with this inhabitation qualification.

3.2 Unlabelled opposition and orientation

The field is prior to a global labelling of its poles. Nothing in its definition calls one pole positive, true, or privileged. A `Polarisation` is additional structure: a dependent section choosing a pole at every opposition. Exchanging every choice gives the opposite polarisation, and exchanging twice restores the first.

This order of explanation matters. Difference is primitive at the field level; orientation is supplied later. The theory can therefore distinguish the fact that a determinable has two poles from the act of taking one pole as the selected outcome in a presentation.

4 Presentation and stratified latency

4.1 Articulation, realization, and resolution

For a presentation p and determination d, write

$$\mathsf{Art}_p(d) \quad \text{when } p \text{ articulates } d,$$

$$\mathsf{Real}_p(d) \quad \text{when } p \text{ realizes } d.$$

The presentation laws require:

(P1) articulation is closed under contrary: $\mathsf{Art}_p(d) \Rightarrow \mathsf{Art}_p(\overline{d})$;

(P2) realization implies articulation: $\mathsf{Real}_p(d) \Rightarrow \mathsf{Art}_p(d)$;

(P3) realization is exclusive: $\mathsf{Real}_p(d) \Rightarrow \neg\mathsf{Real}_p(\overline{d})$;

(P4) each presentation has a fresh trace not articulated there;

(P5) if q articulates $\mathsf{trace}(p)$, it articulates everything that p articulates; and

(P6) each presentation offers an articulated but unresolved opposition.

Resolution is not a further primitive:

$$\mathsf{Res}_p(d) \iff \mathsf{Real}_p(d) \vee \mathsf{Real}_p(\overline{d}).$$

Because contrary determinations share an opposition, $\mathsf{Res}_p(d)$ and $\mathsf{Res}_p(\overline{d})$ are equivalent in effect.

Formal result 4.1 (Resolution orients exactly one pole)**.** *If* $\mathsf{Res}_p(d)$*, then exactly one of* $\mathsf{Real}_p(d)$ *and* $\mathsf{Real}_p(\overline{d})$ *holds.*

Proof sketch. Resolution supplies one of the two realization witnesses. Realization exclusivity rules out the other. In the second case, involutivity converts exclusivity at $\overline{d}$ into exclusion of d. This is `PresentationLayer.resolution_orients`. □

The result is local and typed. It does not assert a global law of excluded middle for arbitrary propositions. It says that when a particular opposition is resolved in a presentation, the presentation realizes one and only one of its poles.

4.2 Four status notions

The formal vocabulary distinguishes:

$$\begin{aligned} \mathsf{Lat}_p(d) &\Longleftrightarrow \neg\mathsf{Real}_p(d), \\ \mathsf{Exc}_p(d) &\Longleftrightarrow \mathsf{Real}_p(\overline{d}), \\ \mathsf{Open}_p(d) &\Longleftrightarrow \mathsf{Art}_p(d) \wedge \neg\mathsf{Res}_p(d), \\ \mathsf{Undrawn}_p(d) &\Longleftrightarrow \neg\mathsf{Art}_p(d). \end{aligned}$$

Status	Articulated?	Realized?	Reading
Realized	yes	d is realized	effective determination
Excluded	yes	$\overline{d}$ is realized	latent through realized contrary
Open	yes	neither pole is realized	drawn but unsettled
Undrawn	no	d cannot be realized	not articulated there

Excluded, open, and undrawn determinations are all latent, as proved by `excluded_latent`, `open_latent`, and `undrawn_latent`. They are not treated as interchangeable. In particular, openness is not ignorance about which pole is secretly realized; it is the non-resolution of an articulated opposition.

Nor does the development prove that these labels form a constructively exhaustive partition of every determination at every presentation. Its positive claim is more precise: along a rooted actual history there are simultaneous witnesses of realized, excluded, open, and undrawn content.

4.3 Offer and trace

Each presentation carries an *offer*, an opposition with some articulated pole for which neither pole is realized. Hence every presentation contains open content. Each presentation also carries a *trace* that is not articulated at that presentation. Hence every presentation contains undrawn content.

Formal result 4.2 (Non-exhaustion at presentation strength)**.** *Every presentation has an open determination and an undrawn determination. Consequently, the thesis that every articulated determination is resolved is inconsistent with the presentation laws.*

Proof sketch. The open witness is extracted from `offer_open`; the undrawn witness is $\mathsf{trace}(p)$ by `trace_fresh`. If every articulated determination were resolved, the offered witness would be both open and resolved. See `openNonExhaustive_of_offer`, `articulativeNonExhaustive`, and `fullResolution_inconsistent_of_offer`. □

This is the first sense in which actuality does not exhaust latency. It is structural, not merely temporal: even before a history is selected, every admissible presentation contains an unresolved articulation and an undrawn trace.

5 Continuation and the archive

5.1 What a continuation does

A continuation $p \xrightarrow{d} q$, formalized by `PresentationLayer.Continues`, settles target d and constrains how q may differ from p. Its clauses say:

(C1) $\mathsf{Real}_q(d)$;

(C2) away from d's opposition, previously articulated realization status is preserved;

(C3) prior articulation persists from p to q;

(C4) q realizes the exact determination $\mathsf{trace}(p)$;

(C5) genuinely new articulation lies at the target or trace opposition;

(C6) away from the target, a newly realized determination is exactly $\mathsf{trace}(p)$;

(C7) $\mathsf{seed}(q) = d$; and

(C8) $\mathsf{promoteNext}(q) = \mathsf{true}$ iff d was not open at p.

The target clause makes the act effective. The off-target clauses express locality: an act cannot silently rewrite unrelated, already drawn content. The articulation clause supplies continuity. The trace clauses produce historical difference. Seed-writing makes the result of an act available to the next act. Cadence records whether the next act should promote open content or turn settled content.

Several apparently stronger properties are derived. Resolution persists through a continuation; a refreshed offer cannot occupy the just-settled opposition; and two successors from one source that settle the same target agree on the realization status of every determination already articulated at the source. These are `Continues.resolution`, `Continues.offer_avoid`, and `Continues.agree_on_drawn`.

5.2 A strict archive

Define

$$p \prec q \quad \Longleftrightarrow \quad (\forall e,\ \mathsf{Art}_p(e) \Rightarrow \mathsf{Art}_q(e)) \wedge \mathsf{Art}_q(\mathsf{trace}(p)).$$

This is `Archive.Precedes`. The first conjunct says that the archive of articulation is monotone. The second says that q contains the trace that was fresh at p.

Formal result 5.1 (Archive advance and finite no-return)**.** *The relation $\prec$ is irreflexive and transitive. Every active transition advances it. Therefore no non-empty finite chain of active transitions returns to its source presentation.*

Proof sketch. Irreflexivity follows because $\mathsf{trace}(p)$ is not articulated at p. Transitivity follows from preservation of articulation. Every active edge inherits the continuation's articulation and trace-carriage clauses, hence advances $\prec$. A positive cycle would imply $p \prec p$, contradicting irreflexivity. These are `Archive.precedes_irrefl`, `Archive.precedes_trans`, `Archive.active_precedes`, and `Archive.no_cycle`. □

The proof reveals the price of irreversibility. The arrow is not obtained from a symmetric state space alone. It is grounded in a principle of historical individuation: each source has content fresh to it, the successor carries that content, and articulation is not deleted. This is a substantive premise, not a theorem of binary opposition.

5.3 An internally marked past

Archive precedence is initially a relation used by the theorist to compare two presentations. The trace-encoding law permits the same order to be recovered from what a later presentation articulates. Define

$$\mathsf{InPast}(q,p) \quad \Longleftrightarrow \quad \mathsf{Art}_q(\mathsf{trace}(p)).$$

In words, p is in the past marked at q when q draws p's trace. Trace encoding says that such a marker is complete enough to carry every articulation of p, not merely a bare name. It does not encode p's realization profile, seed, offer, cadence, or full state.

Formal result 5.2 (Internal past equals archive precedence)**.** *For all presentations p, q,*

$$\mathsf{InPast}(q,p) \quad \Longleftrightarrow \quad p \prec q.$$

Consequently, InPast*, read with the later presentation first, is irreflexive, transitive, and asymmetric.*

Proof sketch. From left to right, trace encoding supplies the monotonic-articulation conjunct of $p \prec q$, while the definition of InPast supplies its trace conjunct. The reverse implication is immediate from precedence. Strict-order properties transfer from the archive, with trace freshness also proving irreflexivity directly. See `Succession.inPast_iff_precedes`, `Succession.inPast_irrefl`, `Succession.inPast_trans`, and `Succession.inPast_asymm`. □

Three strengths must be distinguished. $\mathsf{InPast}(q,p)$ requires only articulation of $\mathsf{trace}(p)$. `ObserverLayer.Registersqp` requires the trace opposition to be resolved, and `ObserverLayer.EffectivelyRegistersqp` requires realization of the exact trace pole. An immediate successor effectively registers its source, and a later endpoint on a positive chain registers the chain source; arbitrary marked past membership alone implies neither stronger condition.

An active edge $p \xrightarrow{d} q$ preserves every past marker already drawn at p and adds p itself to q's past, while p is not in its own past. The theorem does not say that p is the only newly added past member. Along a positive chain from p to q, q resolves p's trace, inherits the past marked at p, and still has an open offer (`Succession.past_settled_future_open`).

The development calls the following package *lived succession.* If a determination o is realized at p, and an act targets an opposition other than o's, then o remains realized at q. At q, the earlier marked past is intact, p is newly included, q is not in its own past, p's trace is resolved, and an opposition remains open:

$$\begin{aligned}\mathsf{Real}_q(o) \ &\wedge (\forall r,\ \mathsf{InPast}(p,r) \Rightarrow \mathsf{InPast}(q,r)) \\ &\wedge \mathsf{InPast}(q,p) \wedge \neg\mathsf{InPast}(q,q) \wedge \mathsf{Res}_q(\mathsf{trace}(p)) \wedge \exists f\, \mathsf{Open}_q(f).\end{aligned}$$

These are `Succession.observer_persists` and `Succession.lived_succession`. Here *observer* means only an effective determination persisting off target; no appearance relation or phenomenology is used.

The result adds a minimal registered-past/open-content contrast. Past marking is strict, the successor is not in its own past, and open content remains at it. No formal present predicate or future presentation is supplied. The direction is derived given trace freshness, articulation-complete trace encoding, articulation preservation, and trace carriage. It is not derived from temporally neutral opposition, and the partial order is not proved linear.

5.4 Recurrence without return

Suppose $p \xrightarrow{\bar{d}} q \xrightarrow{d} r$ are possible acts and $\mathsf{Real}_p(d)$. Then the two acts restore the realization status, at r, of every determination articulated at p:

$$\forall e, \quad \mathsf{Art}_p(e) \Rightarrow \big(\mathsf{Real}_p(e) \leftrightarrow \mathsf{Real}_r(e)\big).$$

At the same time, $p \prec r$. This conjunction is `Archive.recurrence_without_return`.

The recurrence is deliberately restricted. It is status-equivalence on the starting presentation's drawn content, not equality of presentations and not even unrestricted agreement on all determinations. The trace makes the later presentation historically richer. A pattern can recur without the event being numerically the same event.

6 Possibility, plenitude, and excess over selection

6.1 Generative layers

A generative layer G supplies a ternary relation

$$G.\mathsf{Active}(p, q, d),$$

and requires every active triple to satisfy the continuation laws. It is *productive* when every presentation has some active successor:

$$\forall p \ \exists q, d, \quad G.\mathsf{Active}(p, q, d).$$

This is one-step seriality. By itself it neither constructs an infinite history nor selects a coherent successor at every stage.

Presentation structure does not imply productivity. The development defines an inert generative layer with no active transitions. This counterexample, `Possibility.productivity_is_not_presentation_level`, motivates an explicit generative premise.

6.2 The strength of plenitude

The optional plenitude capability combines `Possibility.Rich` and `Possibility.Maximal`.

Richness provides:

(R1) a continuation settling every open determination; and

(R2) a continuation turning the seed at every presentation.

Let $\mathsf{Extends}(G, G')$ mean that every G-active transition is G'-active. Maximality says that if G' extends G, then G also extends G'. Since every generative layer is contained in the full continuation relation, the theorem `Possibility.maximal_iff_extends_full` shows that maximality is extensionally fullness:

$$G \text{ is maximal} \quad \Longleftrightarrow \quad \mathsf{full} \subseteq G.$$

Qualification. Plenitude is therefore a strong name for a strong postulate. It does not follow from opposition or presentation, and it is not merely the absence of a larger chosen possibility relation. It makes every lawful continuation active.

Rich seed-turn continuations transferred by maximality yield productivity (`Possibility.productive_of_plenitude`). The proof is local: for any source, turn its seed.

6.3 Why possibility exceeds every selector

A selector chooses one determination at each presentation and opposition:

$$S(p, o) \in P(o).$$

Its lifted choice on a determination depends only on that determination's opposition. Thus

$$S_p(d) = S_p(\overline{d}).$$

A selector *respects* a generative layer when every active target agrees with its source selection.

Formal result 6.1 (Plenitude excludes universal selection)**.** *If the presentation layer is rich, G is maximal, and at least one presentation is supplied, then no selector is respected by every G-active transition.*

Proof sketch. At the supplied presentation, choose a pole d at the open offered opposition. Openness is symmetric across the two poles. Richness supplies one continuation settling d and another settling $\overline{d}$; maximality transfers both into G. If one selector respected both, it would have to choose both d and $\overline{d}$ at the same presentation and opposition. Coherence says those selector queries are equal, while fixed-point freedom says $d \neq \overline{d}$. See `Possibility.plenitude_excludes_selection`. □

This is modal excess, not contradiction. A rooted system supplies the needed presentation witness. The two poles occur as targets of alternative continuations from one source. They are not co-realized in one presentation. The possible outruns any one orientation because possibility branches across contrary lawful acts.

7 Focus and determinate actuality

7.1 From a selector to a focus

A `Focus` contains a total selector with one constraint: at the seed's opposition, it selects the seed's contrary. A seeded global polarisation is one way to build such a focus, but the dynamic core does not require the selector to be uniform across presentations.

The cadence bit chooses the opposition under consideration:

$$\mathsf{focusOpp}(p) = \begin{cases} \mathsf{offer}(p), & \mathsf{promoteNext}(p) = \mathsf{true}, \\ \mathsf{opp}(\mathsf{seed}(p)), & \mathsf{promoteNext}(p) = \mathsf{false}. \end{cases}$$

The target index is the focus's selection at this opposition:

$$\mathsf{index}_\Phi(p) = \Phi.\mathsf{select}(p, \mathsf{focusOpp}(p)).$$

Consequently, in turn cadence

$$\mathsf{index}_\Phi(p) = \overline{\mathsf{seed}(p)},$$

while in promotion cadence its opposition is the open offer.

Given a possible layer G, actuality is the focused subrelation

$$(\Phi.\mathsf{actual}(G)).\mathsf{Active}(p, q, d) \iff G.\mathsf{Active}(p, q, d) \wedge d = \mathsf{index}_\Phi(p).$$

Actuality does not abolish modal alternatives by declaring them incoherent. It restricts a broader relation by a state-relative orientation.

7.2 What determinacy amounts to

Actuality respects its focus by construction (`Focus.actual_respected`). If two actual edges leave the same source, their targets are equal. Moreover, their successors agree on the realization status of everything articulated at the source.

Formal result 7.1 (Limited actual determinacy)**.** *For actual edges $p \xrightarrow{d} q$ and $p \xrightarrow{d'} q'$,*

$$d = d'$$

and

$$\forall e, \quad \mathsf{Art}_p(e) \Rightarrow \big(\mathsf{Real}_q(e) \leftrightarrow \mathsf{Real}_{q'}(e)\big).$$

Proof sketch. Both active targets equal the single index $\mathsf{index}_\Phi(p)$. With the targets identified, continuation agreement on previously drawn content applies. See `Focus.actual_index_unique`, `Focus.actual_target_determinate`, and the bundled theorem `LatentActualSystem.actual_determinate`. □

The theorem does not prove $q = q'$. New open articulations and other presentation structure may differ within the locality constraints. Actuality is target-determinate without being a unique-future transition system.

Under plenitude the chosen target always has a lawful possible continuation: when promoting, selection preserves openness at the offer; when turning, richness supplies the contrary seed continuation. Thus focused actuality is productive (`Focus.actual_productive`). This dependence matters. Selection alone chooses a target but does not make a matching successor exist.

7.3 Cadence

If an act settles open content, the successor enters turn cadence. If an act turns a realized contrary, the successor enters promotion cadence. Along a history whose seed is effective at turn positions, every actual edge flips the Boolean cadence:

$$\mathsf{promoteNext}(q) = \mathsf{not}(\mathsf{promoteNext}(p)).$$

For a finite actual chain of length n,

$$\mathsf{promoteNext}(q) = \mathsf{after}(n, \mathsf{promoteNext}(p)),$$

where after negates the bit once per act. Hence even-length runs restore the initial cadence and odd-length runs reverse it. These results are `Focus.actual_chain_cadence` and the system-level theorems `actual_chain_cadence_even` and `actual_chain_cadence_odd`.

The Boolean is a minimal control structure, not yet a phenomenology of time. Its philosophical use is to make a structural alternation exact: openness is promoted, settled determination is turned, and the cycle repeats at the level of regime.

7.4 The swirl

Formal result 7.2 (Promotion, reversal, and archive advance)**.** *Suppose*

$$p \xrightarrow{d}_{\text{actual}} q \xrightarrow{e}_{\text{actual}} r$$

and p is in promotion cadence. Then

$$e = \overline{d} \qquad \textit{and} \qquad p \prec r.$$

Proof sketch. The first edge selects a pole of the open offer. Continuation cadence therefore puts q in turn mode and writes $\mathsf{seed}(q) = d$. Focus at a turn selects $\overline{\mathsf{seed}(q)} = \overline{d}$, so the second target is the contrary of the first. Each active edge advances the archive, and transitivity gives $p \prec r$. See `LatentActualSystem.actual_promote_then_reverse`; the parity-indexed run version is `LatentActualSystem.Rooted.actual_chain_swirl`. □

The theorem deserves both emphasis and restraint. It proves an actual one-step reversal after every promotion. It does not prove that one opposition oscillates forever: after the turn, the next promotion concerns the refreshed offer, which is away from the settled seed opposition. Nor does the theorem by itself prove full status recurrence. The stronger `recurrence_without_return` result belongs to a suitable two-edge *possible* sequence and is restricted to previously articulated content.

8 Rooted history and permanent latency

8.1 One effective seed

A rooted system chooses an initial presentation p_0 and assumes

$$\mathsf{Real}_{p_0}(\mathsf{seed}(p_0)).$$

Every active successor realizes its target and writes that target as its seed. The initial fact therefore propagates along every finite actual chain from the root:

$$\mathsf{Reachable}(p) \Rightarrow \mathsf{Real}_p(\mathsf{seed}(p)).$$

This is `LatentActualSystem.reachable_seed_realised`.

The rooted design is premise-reducing. The system does not assume that every presentation, including unreachable ones, realizes its seed. It assumes one effective root and derives the invariant only where actual history reaches.

8.2 The strongest coexistence result

Formal result 8.1 (Permanent latency on rooted actual history)**.** *At every reachable presentation p, there exist determinations witnessing:*

$$\exists d\, \mathsf{Real}_p(d), \qquad \exists e\, \mathsf{Exc}_p(e), \qquad \exists f\, \mathsf{Open}_p(f), \qquad \exists g\, \mathsf{Undrawn}_p(g).$$

Proof sketch. The realized witness is $\mathsf{seed}(p)$, whose effectiveness propagates from the root. Its contrary is excluded. The offered opposition supplies an open witness. The fresh trace supplies an undrawn witness. This is exactly `LatentActualSystem.latency_is_permanent`. □

The conclusion is not that actuality and possibility are vague mixtures. Realization remains exclusive and actual targets remain determinate. Rather, determinacy is local: some content is effective, while contrary, unresolved, and unarticulated content persists around it. Actuality is bounded without being deficient. It is a determination within a field that is not totally resolved.

8.3 The void and the limit of the anti-nullity argument

The development also constructs a `Void.layer`: a coherent presentation layer with an open offer and a fresh trace but no realized determination at all. It proves

$$\exists F, P, p, \quad \forall d,\ \neg \mathsf{Real}_p(d).$$

Thus absolute nullity is compatible with presentation structure. This is `nullity_failure_not_derivable`.

The countermodel blocks an inflated reading of the rooted theorem. The account does not derive something from nothing. It shows that once one seed is effective, the continuation laws preserve non-nullity along its history while never exhausting latency. The origin of effectiveness remains a premise.

9 The concrete Stage model

Abstract implication is not enough to show that all the assumptions can hold together. The `Stage` namespace supplies a concrete model.

Its opposition indices have three forms:

$$\mathsf{base}, \qquad \mathsf{opp}(k), \qquad \mathsf{tr}(m),$$

with Boolean poles. A stage records a base pole, a partial map of settled numbered oppositions, a clock, an independent history counter, its seed, a proof that the seed is effective, and the cadence bit. The current numbered opposition at the clock is the open offer. The current history index is the fresh trace.

The continuation $\mathsf{cont}(s, k, a)$ settles pole a at numbered opposition k, increments the history, advances the clock far enough to refresh the offer, writes the new seed, and computes the next cadence from whether the target was already resolved. The proof `Stage.cont_continues` verifies every locality, preservation, trace, seed, and cadence clause.

The model supplies richness, the full generative layer, a seeded focus, plenitude, and a rooted initial stage. It proves:

- possible and actual productivity;
- incompatibility of plenitudinous possibility with one respected selector;
- an initial actual seed turn;
- a following actual promotion of a genuinely different opposition;
- exact trace locality for new articulation and off-target realization;
- a concrete witness in which an off-target base determination persists while the predecessor enters the successor's non-reflexive past; and
- a possible two-step status recurrence that still advances the archive.

The separate history counter is important. Settling a distant numbered opposition does not silently make a block of unrelated trace tokens effective. Historical novelty is carried one predecessor at a time.

The Stage model is a satisfiability witness, not an empirical cosmology. Its natural numbers and Booleans show that the abstract clauses are jointly inhabited and computationally manageable. They do not establish that physical or lived reality is discrete, binary, or clocked in this way.

10 Metaphysical interpretation

10.1 A process ontology with local determination

The formal picture is processual in a precise sense: presentations are ordered by acts that preserve articulation, settle a target, and generate an irreducible historical trace. A presentation is not merely a timeless set of truths, because its place in the archive depends on what predecessor trace it carries. Yet the process is not flux without identity. Off-target status preservation and monotone articulation provide continuity.

This combination resembles a recurring aspiration of process metaphysics: becoming should produce determinate outcomes without reducing novelty to the rearrangement of a closed inventory. Whitehead's insistence that actual entities are acts of becoming rather than inert substances is a relevant comparison [6]. The present formalization is much thinner. It does not model prehension, causal efficacy, or a theory of relations. Its contribution is the smaller structural claim that local settlement, preserved content, and fresh historical difference can coexist consistently.

10.2 History marked within presentations

The succession layer strengthens the archive's metaphysical role. Without trace encoding, a theorist can prove that $p \prec q$, but the occurrence of $\mathsf{trace}(p)$ within q need not by itself entail all of p's drawn content. With complete encoding, the trace functions as an articulation-complete marker: articulating it is equivalent to standing later in the archive. Historical order is therefore marked in the articulation profiles of later presentations, rather than available only to a metalanguage comparing states. This is extensional marking, not a decoding operation or knowledge predicate.

The theorem named `lived_succession` captures three structural contrasts. There is preservation, because an untouched effective determination and the previously marked past persist. There is non-coincidence, because the successor includes its source but not itself in its marked past. There is continuing openness, because an opposition remains articulated and unresolved. This is a useful minimal pattern of realized content amid marked history and unresolved content; it is not a formal present or a guaranteed future event.

It is not yet lived time in a phenomenological sense. The effective determination called an observer has no perspective or experience, the strict past need not be linearly ordered, and the model has no duration, anticipation, or affect. The result is better understood as a formal precondition for lived succession: a presentation-relative past marker plus asymmetric advance, off-target persistence, and openness. A future account of strands could add the linearity appropriate to a single history without imposing total order on the whole archive.

10.3 Latency is stratified, not a second realm

Latency here is indexed to a presentation. There is no separate warehouse of possible objects. A determination is latent at p exactly when it is not realized there, but the reason for non-realization may differ:

- its contrary is effective;
- its opposition is articulated and still open; or
- it is not drawn into the presentation.

This taxonomy prevents two reductions. Latency is not simply exclusion, since open and undrawn content need no realized contrary. Nor is it simply indeterminacy, since excluded content is determined against. The formalism therefore supports a layered notion of non-actuality without reifying all possibilities as already actual elsewhere.

The account remains extensional: it does not yet distinguish powers, dispositions, tendencies, or degrees of readiness within the broad negative predicate $\mathtt{Lat}_p(d)$. A richer metaphysics of potentiality would need more structure than non-realization and lawful continuation.

10.4 Possibility as lawful transition

Possibility is not interpreted by a Kripke frame of complete worlds. It is the availability of lawful successor acts from a presentation. This makes possibility local and directional. Under plenitude, both poles of an open opposition can head alternative continuations, so no single selection represents all possible acts.

The result offers a precise reading of modal excess:

$$\text{the possible is not one unactualized total description;}$$

$$\text{it is a branching relation that outruns each single orientation.}$$

Actuality, by contrast, is not the elimination of the other branches from the possible relation. It is the focused subrelation. This preserves a clean distinction between modal breadth and actual determination.

The strength of plenitude limits the interpretation. Because maximality is extensionally fullness, the theorem characterizes the consequences of a maximally permissive lawful modality. A weaker modal metaphysics could retain focus, archive, and rooted latency while losing the universal selector-conflict result or productivity.

10.5 Repetition with difference

The distinction between recurrence and return has affinities with philosophical accounts that oppose qualitative repetition to the reinstatement of an identical total state. Bergson's contrast between spatialized succession and duration [2], and Deleuze's account of repetition as productive of difference [3], provide useful points of comparison. No claim of textual equivalence is intended.

The formal contribution is exact:

1. a possible two-act reversal can restore old articulated realization status;
2. actual promotion is followed by contrary realization;
3. in either relevant two-edge case, fresh trace carriage makes the later presentation strictly later in the archive.

Thus recurrence is indexed to a chosen observational profile (`StatusEqOn`), while return would be equality of presentations. The former does not imply the latter. The archive encodes a difference that survives qualitative restoration.

10.6 Individuation and the seed

The seed is the target written by an act into its successor. It is both a record of what has just become effective and the pole available for a future turn. In this narrow sense, the result of one individuation becomes material for the next. This invites comparison with Simondon's emphasis on metastability and the retention of unresolved potentials through individuation [5].

Again, the comparison marks a structural affinity, not a formalization of Simondon. The present seed has no energetic magnitude, incompatibility relation, or transductive domain. What is proved is simpler: settlement writes a state-relative handle that focus can reverse, while a fresh offer keeps another opposition open.

10.7 Internal observation, minimally construed

An optional observer layer relates a presentation, an observer determination, and an observed determination. If something appears to an observer at a presentation, both observer and observed determinations are realized there. Consequently, contrary contents cannot both appear to the same observer there, and open content does not appear as realized.

This gives a minimal immanence condition: observation is internal to the presentation rather than a view from outside the system. It does not yet model perspective, misrepresentation, attention, consciousness, embodiment, or phenomenal character. The dynamic core does not depend on the observer layer.

The succession namespace uses *observer* in a still thinner sense: any realized determination whose opposition an act does not target. Its persistence is derived from off-target preservation and does not require `ObserverLayer` or an appearance witness. The two notions should not be conflated: one constrains appearance, while the other supplies a persisting effective marker through which the structural succession package can be stated.

11 Objections, replies, and limits

O1. *The conclusions are stipulated in the premises.*

Some are. Open offers, fresh traces, trace carriage, locality, and cadence are structural assumptions. The reply is not that Lean makes them self-evident. It is that the formalization turns the complaint into a dependency audit: one can see exactly which conclusions require them. Several burdens are nevertheless reduced. Involution, resolution persistence, offer–seed freshness, refreshed-offer avoidance, cadence parity, and no-cycle are theorems rather than constructor fields.

O2. *Binary opposition is metaphysically reductive.*

The field is a local schema for determinables represented by contrary poles, not a claim that every property or relation in reality is binary. Multi-pole, continuous, or non-oppositional determinables would require another field structure. The present theorem states what follows in the binary case.

O3. *Contrary determination has been confused with logical negation.*

The types prevent this. $\overline{d}$ is a pole exchange within $\mathsf{opp}(d)$. Propositional negation occurs separately in statements such as $\neg\mathsf{Real}_p(d)$. Ontic contrariety and logical denial are related by realization exclusivity, not identified.

O4. *Latency is merely a negative predicate.*

At the broadest level, yes: $\mathsf{Lat}_p(d)$ is non-realization. But the theory refines it into exclusion, openness, and undrawnness with different positive structural conditions. What it does not yet supply is an intensional theory of powers or propensities. That is a genuine extension point.

O5. *Plenitude is too strong.*

Maximality is extensionally the full lawful continuation relation, so the objection is well founded against any attempt to present plenitude as weak. The response is modularity. Plenitude is optional and used for productivity and modal excess over selection. No-return, conditional swirl, determinacy, and much of the archive theory do not require it.

O6. *Focus makes determinism trivial.*

Focus does fix the target by definition. The non-trivial content is what follows when this filter interacts with openness, seed-writing, continuation locality, and cadence. Even then, determinacy is limited: the theory proves a unique target and agreement on old drawn status, not a unique successor presentation.

O7. *The archive simply stipulates the arrow of time.*

The underlying ingredients – fresh trace, articulation-complete trace encoding, carriage, and monotone articulation – are indeed historical-individuation premises. The strict order and its internal marking are derived from them. The theory therefore explains no-return by a precise record principle; it does not derive an arrow from temporally neutral laws.

O8. *An internally marked past is already memory or knowledge.*

It is not. The equivalence between $\mathsf{InPast}(q, p)$ and $p \prec q$ is extensional: q's articulation profile contains an articulation-complete trace marker. There is no decoding function, awareness predicate, or proof that an observer experiences or recalls the order. Nor does mere past membership imply that every marked trace is resolved.

O9. *The non-nullity argument is circular.*

Rooted history assumes one realized seed. The void countermodel proves that this assumption cannot be removed at presentation strength. The valid claim is persistence of actuality from an effective root, not creation of actuality from absolute nullity.

O10. *A Boolean cadence is not lived time.*

Correct. It is the smallest automaton expressing alternation of promotion and turn. The separate `lived_succession` theorem adds internal retention, asymmetric advance, observer persistence off target, and openness ahead. Despite its name, it does not formalize phenomenal duration, variable tempo, anticipation, affect, or embodied temporality. Space also remains outstanding.

O11. *The swirl overstates oscillation.*

Read narrowly, it does not. It proves that every promotion has one immediately following contrary act, with archive advance. It does not prove endless alternation on one opposition. The refreshed offer normally moves the next promotion elsewhere.

O12. *The Stage model is artificial.*

It is artificial in the proper sense of a model construction. Its role is to witness joint satisfiability and catch inconsistent axiom packages, not to supply empirical evidence or a unique intended interpretation.

O13. *The observer theory does not explain experience.*

It does not. The optional appearance layer proves internal co-realization and some exclusivity consequences. The succession theorem separately proves off-target persistence for a realized determination called an observer. Neither supplies consciousness, error, salience, or phenomenal character.

12 Conclusion

The latent–actual formalization gives a conditional answer to a difficult structural question. A world of unlabelled oppositions can support locally oriented realization without global closure. Lawful acts can preserve what is drawn while making a target and a fresh trace effective. A maximally rich possible relation, when supplied a presentation, can exceed every single selector, while focus carves out a productive and target-determinate actuality. From one effective root, realized, excluded, open, and undrawn content coexist throughout reachable history. Trace encoding marks archive precedence inside later articulation profiles. Promotion leads to reversal, but trace carriage prevents return.

The strongest philosophical lesson is methodological as much as ontological. Determinacy need not mean exhaustiveness; recurrence need not mean identity; and conditional metaphysical claims become clearer when their generative, modal, and historical premises are separately named. The void countermodel keeps the account honest: presentation does not make itself actual. The Stage model shows, conversely, that actuality, permanent latency, modal excess, focused selection, and archival irreversibility can inhabit one coherent structure.

What remains is substantial. The binary schema must be generalized if it is to address continuous or plural determination. Productivity must be distinguished from completed infinite succession. The development now marks a strict past within presentations and proves a limited persistence package named lived succession, but metric or continuous time, phenomenal duration, and a formal future relation remain absent. So do space, causation, embodiment, and full-blooded observation. These are not small omissions, but the formal core gives future extensions a disciplined base: new premises can be added, and their real consequences can be proved rather than suggested.

A Argument-to-theorem map

Lean declaration	Role in the argument
`Field.neg_involutive` and `Field.neg_free`	Contrary exchange is involutive and has no fixed point.
`PresentationLayer.resolution_orients`	A resolved opposition realizes exactly one pole.
`PresentationLayer.openNonExhaustive_of_offer`	Every presentation contains an open determination.
`PresentationLayer.articulativeNonExhaustive`	Every presentation contains an undrawn determination, namely its fresh trace.
`PresentationLayer.fullResolution_inconsistent_of_offer`	Universal resolution of articulated content is inconsistent with the open offer.
`PresentationLayer.trace_encodes`	Articulating a source trace carries every determination articulated at that source; it is not a complete state encoding.
`Continues.resolution`	Resolution persists through lawful continuation.
`Continues.newly_articulated_is_trace` and `Continues.off_target_novel_is_trace`	Transition locality confines non-target novelty to the predecessor trace.
`Archive.no_cycle`	No non-empty finite active chain returns to its identical source.

Lean declaration	Role in the argument
`Succession.inPast_iff_precedes` and `Succession.inPast_asymm`	A presentation's past markers are exactly archive predecessors and form an asymmetric relation.
`Succession.past_grows_strictly`	Every active edge preserves earlier past markers and adds at least its source.
`Succession.past_settled_future_open`	A positive chain resolves its starting trace, preserves its starting past, and ends with an open offer.
`Succession.observer_persists` and `LatentActualSystem.lived_succession`	Off-target realized content persists through one act, conjoined with strict past growth, source-trace settlement, and continuing openness.
`Archive.recurrence_without_return`	A two-step possible reversal restores old drawn status while the archive advances.
`Possibility.productive_of_plenitude`	Richness plus maximality gives a successor at every source.
`Possibility.plenitude_excludes_selection`	No one selector agrees with both contrary branches made active by plenitude.
`Focus.actual_respected`	Focused actuality agrees with its selector.
`Focus.actual_productive`	Plenitude supplies a possible continuation matching every focused index.
`LatentActualSystem.actual_determinate`	Actual edges from one source share a target and agree on previously drawn realization status.
`LatentActualSystem.reachable_seed_realised`	One effective root seed propagates along actual reachability.
`LatentActualSystem.latency_is_permanent`	Realized, excluded, open, and undrawn witnesses coexist at every reachable stage.
`LatentActualSystem.actual_chain_cadence_even` and `LatentActualSystem.actual_chain_cadence_odd`	Actual cadence is determined by finite run parity.
`LatentActualSystem.actual_promote_then_reverse`	Every promotion is followed by contrary realization and archive advance.
`LatentActualSystem.Rooted.actual_chain_swirl`	The swirl holds at every promoting parity position in a rooted actual run.
`nullity_failure_not_derivable`	Presentation structure alone permits absolute nullity.
`Stage.actual_productive`, `Stage.actual_novelty`, `Stage.lived_succession_witness`, and `Stage.oscillation_possible`	The concrete model witnesses productivity, novelty, marked succession, and possible recurrence without return.

B Premise ledger

Layer	Supplied structure	Principal derived consequences
Field	Pole exchange, fixed-point freedom, two-pole dichotomy	Involution; equality-or-contrary within a fibre
Presentation	Articulation closure, realization exclusivity, open offer, fresh trace, articulation-complete trace encoding	Exact orientation of resolution; open and undrawn non-exhaustion; internally marked archive order
Continuation	Target realization, preservation, trace carriage, locality, seed-writing, cadence equation	Resolution persistence; offer avoidance; archive advance; local agreement
Plenitude	Open continuations, seed turns, extensional maximality	Possible and actual productivity; no universally respected selector
Focus	Total state-relative selection; contrary selection at seed opposition	Unique actual target; promotion/turn index; actual focus-respect
Root	One initially realized seed	Reachable seed effectiveness; turn cadence; reachable non-nullity and permanent latency

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